

DeepSeek V3 → V4: What Changed & What Matters

Efficiency unlocks capability. The 1M-context jump is architectural, not scaled.

CLIFFFORD
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v1.0 · Internal

2.4×

Parameter scale jump
671B → 1.6T

8×

Context length jump
128K → 1M

10%

KV cache vs. V3.2
90% reduction

\$5.6M

V3 training cost
V4: still open-source

V3 proved constraint drives innovation. V4 proves efficiency unlocks capability. The jump from 128K to 1M context is architectural: hybrid sparse attention (CSA + DSA) cuts KV-cache costs by 90%, making long context a default, not a premium.

Head-to-Head the architecture delta

Dimension	V3 (Dec 2024)	V4 (Apr 2026)
Scale	671B / 37B active	1.6T / 49B active (Pro)
Context	128K tokens	1M tokens — default on both
Attention	MLA (latent KV compression)	CSA + DSA (hybrid sparse)
Precision	FP8 mixed	FP4 + FP8 (experts FP4, rest FP8)
Optimizer	AdamW variants	Muon — per-param update scaling
Residual	Standard hyper-connections	mHC (Manifold-Constrained)
Inference	Baseline open MoE	27% FLOPs, 10% KV vs. V3.2 at 1M
Variants	Single model	Pro + Flash (1.6T / 284T)

The Strategic Shift what V4 signals about DeepSeek's direction

- **From efficiency to viability.** V3 showed you could train cheap. V4 shows you can serve 1M context at prices undercutting 128K competitors by an order of magnitude.
- **Dual-model strategy.** Pro targets ceiling; Flash targets speed/cost. Not "flagship + cut-down" — Flash is separately trained and competitive on most benchmarks.
- **Agent-first positioning.** V4 is explicitly optimized for agentic coding (Claude Code, OpenClaw, OpenCode). The model is infrastructure for autonomous workflows, not just chat.
- **Long context as default.** 1M is not a premium tier. Both variants ship it standard. The competitive axis shifts from "can it?" to "how much does it cost?"

The real delta is not the parameter count — it is the cost curve. V3 was a proof that Chinese labs under export controls could out-engineer unlimited-budget Western competitors. V4 is a proof that once you solve efficiency, capability becomes a commodity. When 1M context costs less than 128K used to, every application redesigns around it.